

A stronger MARL thesis

Research direction, code assessment,
and a concrete execution plan

Prepared for **Akshat Guduru**
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Recommendation

Keep the framework. Focus the thesis on the difference between useful information, learned message dependence, and coordination under communication deadlines and failures.

Includes a pinned GitHub review, completed-result assessment, recent MARL literature, Penn research fit, seven ranked directions, a detailed experiment plan, and reading priorities.

Repository: [akki-g/marl-comm](#)
Commit: [ea5bf44d51c59f029a5b8f4da39fce0a73b1a51](#)

Existing numerical findings are attributed to repository records. New experiments and contributions are proposals, not completed results.

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Reading guide

Start with Sections 1-2 for the decision and evidence. Section 5 tests novelty against the closest papers. Sections 7-9 specify the recommended project. Section 10 explains the pivots; Section 11 turns the recommendation into a schedule.

1. The decision I recommend

Keep the framework. Change the scientific center of the thesis. Do not launch another broad communication-mechanism sweep yet. Your best next move is a focused extension from “which communication module helps on which task?” to **separating available information, learned dependence on messages, and useful coordination under a deadline and unreliable communication.**

A working title is **“When Does Communication Help? Information, Deadlines, and Failure in Cooperative Multi-Agent Learning.”** A more technical title, once the evidence supports it, is **“Separating Communication Capability and Learned Reliance in Decentralized Robot Teams.”** These are proposed research directions, not claims that the contributions have already been established.

Your current work is stronger as an experimental platform than your description suggests. It already includes matched-capacity controls, frozen experimental protocols, message interventions, action diagnostics, and a substantial record of completed experiments. The main weakness is that the tasks and results do not yet isolate the cause your thesis wants to explain. In particular, communication changing an agent's behavior, or hurting performance when removed, does not prove that communication is necessary for the task.

My recommendation has three layers:

1. **Minimum viable thesis:** a rigorous study of when learned message reliance agrees or disagrees with the actual information requirements of a small, explicitly defined task family. Preserve your current negative results as motivation and historical evidence.
2. **Preferred paper:** add a robotics-relevant task, recurrent local controls, realistic delivery timing, and held-out communication conditions. Establish a useful empirical finding that survives these controls.
3. **Optional method contribution:** introduce one small intervention only after identifying a reproducible failure that existing baselines do not resolve. Do not start by inventing another attention mechanism.

Your repository already separates learning benefit from frozen-policy reliance and already attempts private-information diagnostics. The new work proposed here is the calibrated task-level reference, validated recurrent controls, and explicit communication timing.

The central candidate contribution is a **calibrated experimental account of the gap between what information makes possible and what trained agents actually exploit**, under controlled timing and sensing conditions. That is more specific than a communication ablation study, but its novelty remains conditional. Several 2026 papers overlap with parts of this idea. A two-week comparison against those papers should precede months of implementation.

If you strongly prefer a larger pivot, my second choice is **resilient multi-robot information gathering under intermittent connectivity**, and my third is **generalization to changing team composition**. I would not make a foundation-model, LLM-critic, or general-purpose world-model project the primary thesis this late in the existing development.

This recommendation is about producing credible research, not guaranteeing publication or admission. A clear question, an honest negative finding, a convincing explanation, and an advisor who can describe your independent contribution are more useful than a fashionable title with weak evidence.

2. What I reviewed, and what the evidence establishes

I inspected the current GitHub repository through the GitHub connection, obtained a local checkout at the commit above, read the current implementation and tests, and reconciled historical result documents with the latest September 14 entries in `agents.md`. The review includes the actor/communication interface, channel semantics, task observations, PCP and MAPDN adapters, analysis tools, experiment protocols, and current experiment summaries.

Three evidence levels matter:

- **Directly inspected code:** the implementation supports the architectural findings in this report. A small isolated reproduction confirmed the current serialized-path bug described below.
- **Repository-reported experiments:** numerical results below come from tracked documentation and the development log. I did not independently rerun the scientific experiments or recalculate them from raw checkpoints and trajectories.
- **Proposed research:** all new experiments, estimates, methods, schedules, and acceptance criteria in this report are recommendations. They have not been executed.

The repository ignores `runs/` and `results/`. Many links in the tracked documentation therefore point to evidence that is absent from a fresh GitHub checkout. The development log reports substantial raw-data verification, but those reports are not a substitute for independently accessing the original artifacts. The review did not have your Newton filesystem or the local historical evidence archives. No training, dependency upgrade, repository modification, push, or faculty contact was performed.

[Reviewed GitHub snapshot](#) · [Current development log](#) · [Artifact exclusion rules](#)

2.1 The current results change the next step

Study	What the repository reports	Appropriate interpretation
Simple Spread V2	231 planned runs, 224 completed, about 43 V100 GPU-hours; main comparison has five seeds per method	You have already done a substantial mechanism/ablation study. Repeating a larger version is unlikely to solve the conceptual problem.
Simple Spread main comparison	Broadcast final return -462.4 versus Identity -474.3; paired difference +11.9, descriptive 95% interval [9.6, 14.4]	A modest fixed-budget benefit in this setup; not proof of task-level necessity or asymptotic superiority.
PCP D4	40 policies: two prey-visibility conditions, four controls/methods, five training seeds	The controlled follow-up is complete according to the latest log, even though older status documents still describe it as incomplete.
PCP primary contrast	Restricted-visibility K3 Broadcast minus LocalCapacity contact-by-50: -28.90625 percentage points; interval [-63.515625, +15.390625]	No demonstrated positive Broadcast advantage; considerable uncertainty. This does not establish equivalence or universal harm.
PCP reliance	Restricted-visibility live minus severed contact: -0.859375 percentage points; interval crosses zero	Removing messages does not reveal reliable useful reliance in this tested setting.
PCP supplementary interaction	Restricted-minus-global reliance: -5.859375 percentage points; interval [-11.171875, -2.03125]	Opposite to the expected ordering, but supplementary/post-hoc. It should not be relabeled as the original primary test.
Longer PCP local baselines	Four fresh Identity policies at 1.2M frames learn, but fail the declared late stability criterion	Extending the budget did not establish convergence. It also did not test a new communication effect.
MAPDN September 14 comparison	15 policies at 32,768 frames; 480 live/severed evaluation trajectories	The bounded comparison is complete according to the log. The allotted training budget remains an unresolved limitation.
MAPDN learning effect	LocalCapacity minus Broadcast mean voltage violation: -0.0002040514 p.u.; interval [-0.0003893493, +0.0000132082]	With this sign convention, the point estimate favors LocalCapacity; the interval crosses zero.
MAPDN reliance	Severed minus live voltage violation: +0.0001173408 p.u.; interval [-0.0001506609, +0.0004298073]	The mean suggests useful reliance, but uncertainty is large and the direction varies across seeds.

All quoted intervals above are repository-reported descriptive seed-level intervals. Their existence should not be turned into a sweeping significance claim, especially after many comparisons. [Simple Spread results](#) · [Latest completed PCP/MAPDN record](#)

The private-information diagnostic is equally important. PCP produced exact observation twins in some constructed cases, but all tested native reward/contact rollouts were zero over the chosen horizon. Its positive distance-to-prey result is a secondary approach diagnostic, not a rescue of the native objective. The

MAPDN interventions changed the receiver's local voltage measurements, so they were not exact observation twins. Neither experiment established useful private information on its intended native endpoint. [Diagnostic design](#) · [Recorded outcomes](#)

The implication: you should not spend the next month searching for a communication module that finally wins. First establish that the chosen task exposes a decision-relevant information bottleneck, and distinguish failure to learn from absence of useful information. The present negative findings are a reason to sharpen the experiment, not erase the history.

2.2 What is worth retaining

The reusable core is substantial:

- A BenchMARL/MAPPO backbone with communication isolated in the actor path.
- A parameterless Identity control and an active LocalCapacity residual control.
- Broadcast, gated, attention, and graph communication implementations.
- Strict actor-input contracts and a separately constructed PCP critic state.
- Realized communication-dropout masks retained for PPO recomputation.
- Paired live/severed evaluation and exact null checks for local controls.
- Domain metrics, checkpoint provenance, seed bookkeeping, and failure records.
- A real MAPDN integration with chronological data partitions and physical outcome reporting.

These are research assets. Rewriting them in a new MARL framework would consume time without automatically producing a new scientific contribution. [Actor implementation](#) · [Communication implementations](#) · [Analysis code](#)

2.3 The code and presentation gaps that matter scientifically

Priority	Finding	Why it matters	Next action
P0	The current actor comparison is feedforward; the existing diagnostics do not supply a recurrent local comparison	Messages can substitute for missing memory, so an apparent communication advantage may be a memory comparison	Add a properly trained recurrent local control before making a necessity claim; verify sequence reset and replay.
P0	RESULTS_v2_simple_spread.md calls the task fully observable while listing observations that omit other-agent velocities	Seeing all positions and landmarks is not full physical-state observability	Correct the claim and retain the original report as history. Call this a broad position-observation reference, not a full-state null.
P0	The current next-phase launcher passes serialized string paths to a reader expecting Path	A documented evaluation failure remains reproducible in current source	Normalize paths in the new source version and smoke-test the serialized training-closure-to-evaluation route. Preserve the historical workaround and results.
P1	Graph support does not currently make MAPDN's feeder topology the actor's communication graph	A generic graph layer is not evidence of electrical-topology-aware coordination	Define the communication network separately from the electrical network; wire and validate the chosen masks explicitly.
P1	Learned top-K selection is an idealized global-score scheduling abstraction	Acquiring every candidate's keys/scores can require uncharged communication	Keep it as an explicitly labeled centralized scheduling upper reference, or implement local admission and count control traffic.
P1	Current channel support does not include real inter-step queues, message age, burst outages, or reconnection	Independent sender dropout does not represent these deployment conditions	Add only the channel semantics needed by the selected research question.
P1	The random-K option is a seeded fixed priority subset	It is not the same baseline as fresh random scheduling every step	Name the baseline accurately; add periodic and iid schedules only where relevant.
P1	Important results are scattered across historical documentation and an extremely long log	Reviewers cannot readily determine what is current or reproduce the figures	Create a compact current evidence index with immutable artifact references and explicit historical labels.
P2	Role embeddings/configuration options exist, but current PCP does not instantiate the older detector/capturer task semantics	Your old heterogeneous PCP description does not describe the current Simple-Tag-derived task	Describe the implemented task exactly; treat heterogeneous roles as a new task version.

The path bug was checked with an isolated standard-library reproduction of the actual reader: Path input succeeds and the corresponding string raises `AttributeError` at `.exists()`. This was not a rerun of the full training pipeline. [Launcher](#) · [Reader](#)

The graph, channel, and scheduler limitations are mostly scope limitations, not hidden implementation bugs. The repository already documents the idealized scheduling assumption. Report them accurately rather than treating every missing deployment feature as defective software. [Graph module](#) · [Channel module](#) · [Communication methods and assumptions](#)

3. Clarifying the Penn target

The program name you likely mean is **Applied Mathematics and Computational Science (AMCS)**. However, Enyioha's own biography says he completed his PhD in **Electrical and Systems Engineering**, supervised by **George Pappas and Ali Jadbabaie**, and was affiliated with **GRASP and PRECISE**. If your goal is specifically his research lineage, Pappas/ESE is the most direct verified connection. His current stated interests also closely match communication-constrained distributed control and learning. [Enyioha's biography and research statement](#)

Penn's GRASP page says doctoral students generally apply through CIS, ESE, or MEAM, and that there is **no separate Robotics PhD degree**. The Robotics master's is a distinct program. AMCS is not incompatible with robotics: Pratik Chaudhari's GRASP profile lists AMCS doctoral students and alumni. The exact appropriate route depends on the advisor and intellectual emphasis; current AMCS admissions details and advisor eligibility were not fully verified in this review. [GRASP doctoral study](#) · [Robotics master's](#) · [Chaudhari's group](#)

Research match	Why the fit is credible	What your thesis should demonstrate
George Pappas / ESE / GRASP	Networked systems, control, robotics, learning-enabled autonomy; verified advisor lineage	A precise information/control question, rigorous evaluation, and explicit failure assumptions.
Alejandro Ribeiro / Alelab / ESE	Graph learning and distributed collaborative intelligence	Meaningful topology or scale generalization, with communication costs and locality accounted for.
Vijay Kumar / GRASP	Cooperating robot teams and the interaction of control, perception, and communication	A physical coordination objective, realistic communication disruptions, and convincing robot simulation or demonstration.
M. Ani Hsieh / ScalAR / MEAM	Distributed sensing, heterogeneous teams, and robot-environment dynamics	A strong information-gathering or monitoring task with interpretable environmental consequences.
Pratik Chaudhari / GRASP	Learning, control, optimization, and mathematical foundations	A focused generalization or representation question with a clear mathematical/statistical argument.

This is my research-fit assessment, not a statement that any faculty member is recruiting or that a particular project improves admission probability. [Pappas](#) · [Alelab](#) · [Kumar](#) · [Hsieh](#) · [Chaudhari](#)

There is a concrete recent connection: **Long-Term Mapping of the Douro River Plume with Multi-Agent Reinforcement Learning**, coauthored by Pappas and accepted at ICRA 2026, studies communication/energy-efficient AUV monitoring with intermittent central coordination. Your possible decentralized study would have a different execution model, but the topic remains directly relevant. Penn's **MOCHA** work, published at ICRA 2024, supplies a useful systems precedent for opportunistic peer communication in heterogeneous robot teams. [Douro paper](#) · [MOCHA paper](#) · [MOCHA author repository](#)

The practical step is to ask Enyioha which current advisor and program best fit the research you want to do. Do not optimize the thesis around an assumed department name or a lab connection alone.

4. Where the MARL field is active now

The following is a research map based on representative primary papers available by September 19, 2026. “Active” means visible recent research activity in the inspected literature, not a bibliometric ranking of the entire field.

4.1 Useful, efficient, and realistic communication

The field has moved well beyond demonstrating that a differentiable message improves reward. Recent work considers recipient selection, communication credit assignment, explicit bandwidth limits, and delay/congestion. For your experience, this remains the best intersection of topical relevance, code reuse, advisor fit, and finishability. It is also crowded: a simple value gate, attention layer, or bandwidth penalty is unlikely to be a distinctive contribution. [SCoUT](#) · [SLIM](#) · [CAIC](#)

4.2 Generalization beyond the training team and environment

An important question is whether agents cooperate when partners, team size, geometry, roles, or tasks change. This is more meaningful than evaluating only unseen random seeds on the same task distribution. **HyperMARL** addresses flexible agent diversity through hypernetworks, while **Symmetries-enhanced MARL** uses structured models for distributed swarming. Your graph/attention background transfers, but success requires explicit held-out structural splits. [HyperMARL, NeurIPS 2025](#) · [Symmetries-enhanced MARL, 2025 preprint in the inspected record](#)

4.3 Safe and resilient decentralized control

This is particularly relevant to robotics: achieving a task while respecting collision or state constraints, including when neighbors or messages disappear. **DGPPO** and **GCBF+** are strong recent anchors. The challenge is to distinguish an empirical collision rate from a mathematical guarantee. A learned communication policy plus a safety penalty does not automatically provide safety. This direction fits Penn well, but adds control-theory and verification demands to your current implementation work. [DGPPO, ICLR 2025](#) · [GCBF+, IEEE Transactions on Robotics 2025](#)

4.4 Offline, data-efficient, and continual multi-agent learning

Offline MARL learns from fixed datasets; continual MARL asks how teams adapt across changing tasks without repeatedly starting over. **Oryx** and **MEAL** are useful entry points. These could exploit your statistics background, but data coverage, teammate nonstationarity, and evaluation protocol become the main research problems. Existing trajectories are not automatically an adequate offline dataset, and a new framework may be needed for a faithful baseline. [Oryx, NeurIPS 2025](#) · [MEAL, ICML 2026](#)

4.5 Foundation models, world models, and language-mediated coordination

These topics attract substantial attention, but “multiple LLM agents” and MARL are not interchangeable. A language-agent system can coordinate without learning a multi-agent policy through reinforcement learning. A world model also introduces model error and potentially substantial training costs. One recent embodied-agent study reports that dialogue can reduce conflicts while lowering task success, illustrating why visible interaction is not enough. For you, a frozen high-level language planner paired with existing low-level policies could become a side project; training a foundation model or replacing the critic with a large LLM is a poor default thesis pivot. [Embodied coordination through world-model dialogue, May 2026 preprint](#)

4.6 Better benchmarks and evaluation

This is not a consolation category. It asks whether apparently successful agents solve the intended problem or exploit shortcuts. **Probing Dec-POMDP Reasoning in Cooperative MARL** audits 37 scenarios and separates several forms of information dependence and coordination. A carefully designed benchmark or diagnostic can contribute substantially, but your work must go beyond a generic claim that common benchmarks are insufficient. [AAMAS 2026 paper and author project](#)

For your thesis, the most promising intersection is **communication + rigorous task design + deployment sensitivity**, with one clear robotic consequence. The largest trend in the field need not be your best project.

5. The nearest prior work: what you cannot claim as new

This is the most important reading section. The remaining gap must survive these comparisons. Preprint status is not a reason to ignore overlapping ideas, and a venue name in an arXiv comments field is not by itself verified acceptance.

Work and status	Existing contribution relevant to you	Consequence for your proposed thesis
On the Pitfalls of Measuring Emergent Communication , AAMAS 2019	Warns that apparent signaling or correlations do not establish useful communication	Do not treat message activity, attention patterns, or action influence alone as utility.
Learning Individually Inferred Communication for Multi-Agent Cooperation / I2C, NeurIPS 2020	Uses causal influence to identify relevant communication partners	“Learn whom it is necessary to communicate with” is already established territory.

Work and status	Existing contribution relevant to you	Consequence for your proposed thesis
Efficient Multi-Agent Communication via Shapley Message Value, IJCAI 2022	Assigns communication value using cooperative-game reasoning	Shapley or marginal-message scoring is not an untouched thesis direction.
DACOM: Learning Delay-Aware Communication for Multi-Agent Reinforcement Learning, 2022 preprint record	Explicitly addresses delayed communication	Adding a delay parameter alone is insufficient novelty.
IMGS-MAC, 2022 preprint / subsequent IEEE publication	Causal message analysis and communication-budget reduction	Lossless/minimal empirical budgets need a more specific distinction.
Probing Dec-POMDP Reasoning, AAMAS 2026	Diagnoses history, teammate information, and coordination in learned policies	A generic diagnostic audit would overlap; distinguish statistical probes, interventions, and a certified task subcase.
SCoUT, March 2026 preprint	Counterfactual communication advantages for send/recipient decisions	Leave-one-message-out utility gating is already proposed.
SeqComm-DFL, April 2026 preprint, revised May	Decision-focused sequential communication and information/necessity analysis	Neither decision-aware messaging nor a basic communication-necessity proposition is enough by itself.
SLIM, May 2026 preprint	Separates policy latent capacity from communication bandwidth and defines a unified budget	Your capacity controls are essential methodology, but not a first contribution. Timing/network effects are a possible extension to examine.
COBE, AAMAS 2026	Communication selection balancing long-term value and cost	A generic cost-benefit or Pareto-knee method is crowded.
CoDe, AAAI 2025	Aligns message intent and timeliness under fixed and time-varying delays	Timestamps, recency weighting, and delay-tolerant fusion already have direct predecessors.
CAIC, UAI 2026	Models shared-channel queueing and uses delay-aware intent communication	Stale messages being worse than silence and adaptive frequency are already reported.

The prospective distinction is narrower: **a task family in which information availability and time-to-use are independently controlled, one subcase has an explicit capability bound, trained-policy reliance is measured separately, and failure distributions are varied without changing their mean budget.** I did not identify an inspected paper evaluating this exact combination. That does not establish that the scientific claim is new; the proposed study still needs a distinct unresolved question and finding.

A combination of familiar ingredients can still be a weak paper if it yields only expected plots. The contribution becomes stronger if it exposes a reproducible failure of an existing method or a common evaluation assumption, explains the mechanism, and yields a useful design decision. Examples include showing that a bandwidth-efficient policy fails specifically when its information arrives after a decision boundary, or that a method's apparent necessity disappears after adding local history.

Do not claim that these outcomes will occur. Write down competing predictions and allow the data to reject the intended story.

6. Your options, ranked

These rankings are my judgment given your current code, CS/statistics/mathematics background, advisor, and existing cluster experience. Time estimates assume roughly 10-15 focused research hours per week, dependable access to the existing compute, and no long hardware procurement. They are planning ranges, not measured guarantees.

Rank	Direction	Code reuse	New learning burden	Main novelty risk	First credible evidence
1	Calibrated communication utility under deadlines and failures	High	Moderate: recurrent evaluation and channel timing	Closely adjacent 2026 work; task must yield a distinct finding	2-4 weeks for diagnostic results; 10-14 weeks for a coherent draft
2	Resilient cooperative sensing / information gathering	Moderate to high	Moderate-high: estimation and physical task design	Generic MARL monitoring is already studied	4-6 weeks for a pilot; 12-18 weeks for a paper
3	Generalization to new team composition / failed teammates	Moderate to high	Moderate: variable teams, partner splits	"GNNs generalize to more agents" is established	3-5 weeks for a pilot; 10-16 weeks for a paper
4	Safe communication-aware navigation/control	Medium	High: control constraints and guarantee assumptions	Adding a barrier filter or safety reward alone	5-8 weeks for a pilot; 14-22 weeks for a careful paper
5	Topology-aware power-grid coordination under communication constraints	High adapter reuse	High domain validation burden	Feeder-specific findings and weak private-information identification	4-6 weeks for a pilot; 12-18 weeks for a paper
6	Offline or continual cooperative learning	Low to moderate	High: datasets, new evaluation protocols	Strong baselines and distribution shift are difficult	5-8 weeks for a pilot; 14-22 weeks for a paper
7	LLMs or world models for MARL	Low	High, with uncertain compute cost	Fashionable integration without a clear learning contribution	Hard to bound reliably from current evidence

Why option 1 wins: it turns a weakness already exposed by your experiments into the research question, preserves most of your infrastructure, suits your advisor's expertise, and allows a rigorous contribution without needing a new simulator stack or large model. **Why option 2 is the best larger pivot:** it gives communication a concrete physical purpose while remaining close to distributed control and Penn robotics.

7. The recommended project, specified concretely

7.1 One primary research question

When remote information is independently known to be useful, how much of a learned team's failure under delayed or missing communication comes from information that is no longer available, and how much comes from an avoidable failure to use the information that remains?

This question changes what you measure. A decline under message loss is unsurprising. A more useful result measures that decline relative to an achievable information-limited reference and identifies whether memory, a simple fallback, or a particular communication design closes the remaining gap.

Use three competing hypotheses:

- **H1: information limitation dominates.** Learned policies track the attainable reference as timely delivery falls. There is little algorithmic headroom; the main design recommendation concerns sensing or channel provision.
- **H2: learned dependence dominates.** Performance falls substantially faster than the reference because agents over-trust missing/stale messages or fail to use local history. A simple robust baseline may outperform a more elaborate learned communicator.
- **H3: the apparent communication need is recoverable locally.** A recurrent local policy closes much of the feedforward gap in a task where local history contains the relevant information. The correct conclusion concerns the tested information/memory regime, not all decentralized control.

These hypotheses must be evaluated in separate, explicitly defined information regimes. Do not use a toy in which hidden information is impossible to infer locally to “test” whether local memory can infer it. Include a locally recoverable regime if memory substitution is part of the question.

7.2 Task ladder: one calibration problem, one robot problem

Level A: a small signal-and-decision problem. A scout observes a random mode or correct goal; a responder must commit to a decision before a deadline. The mode is independent of all responder information before the decision. The experiment controls whether a message arrives in time. This is a calibration fixture with a known limit, not the headline robotics contribution.

Level B: a VMAS scout/responder task with actual movement. A scout can observe which of two sites requires service. A responder must choose a corridor or allocate itself to a site before a commitment boundary. The scout's observation is private until communicated; its visible motion must not reveal the answer in the certified version. Navigation dynamics, travel time, obstacles, and control effort remain meaningful. Report correct assignment separately from reaching the target, so navigation incompetence is not misdiagnosed as communication failure.

Use two related variants of the same physical task:

Variant	Information structure	Scientific role
Private cue	Only scout knows the current goal before commitment; receiver history is independent of it	Establishes a controlled explicit-communication advantage under stated assumptions.
Redundant or recoverable cue	Responder receives the cue locally, perhaps briefly before a memory interval	Tests whether the same architecture relies on messages unnecessarily, and whether local history removes the gap.

Keep dynamics, goal prior, action costs, horizon, team size, and reward identical wherever possible. Change the observation channel deliberately. If an observation change also changes task difficulty, report that fact and retain an oracle-information baseline in both settings.

Level C, optional: one existing task such as PCP provides an external check. It will not inherit the calibration proof. Use it to test whether the qualitative failure mechanism persists, with bounded claims. MAPDN can remain in an appendix or a separate application paper; forcing three domains into the main argument will dilute the first paper.

7.3 An elementary calibration argument

Let a fair hidden bit Z determine the correct responder action. Suppose that, before the commitment time, the responder's entire available history H is independent of Z . That history includes observations, previous actions, rewards, clock information, agent identities, shared randomness, and all possible physical signals. Then any local policy, including a recurrent policy, succeeds with probability at most $1/2$.

The proof is short: conditional on any available history and independent policy randomness, both values of Z remain equally likely. Choosing either action therefore has expected success $1/2$; averaging over histories does not improve it. Centralized training cannot reveal a newly sampled execution-time bit that is absent from the actor's history.

With a correct one-bit message delivered before commitment, perfect decision success is possible in this calibration problem. If an exogenous delivery event is sampled independently of the joint pair (Z, H) , with no content-dependent timing or selection, arrives in time with probability q , and reveals Z whenever it arrives, the attainable decision success is:

$$S^*(q) = q + (1 - q)/2 = (1 + q)/2.$$

This is a **sanity-check bound**, not a novel theorem. The continuous robot version can have additional navigation errors and may not attain this curve. Distinguish its information-limited decision component from full-task success.

The assumptions matter. A sender that chooses when to transmit can encode information in silence or timing; an exogenous q formula cannot be applied unchanged. A moving scout can signal through its trajectory. Shared rewards before the decision can leak the answer. A fixed episode list, correlated reset positions, or repeated training seeds can expose the mode. All are part of the information contract, not minor implementation details.

A changing binary mode offers another transparent reference: if it flips independently with probability r per step, a correct observation d steps old predicts its current value with probability $[1 + (1 - 2r)^d]/2$ for $0 \leq r \leq 1/2$. This elementary calculation can calibrate freshness sensitivity, but it is not a substitute for

comparison with CoDe, CAIC, or existing age-of-information work. Start with the static-bit deadline task; add the changing mode only if it answers a distinct question.

7.4 Four outcomes, not one saliency score

1. **Information-limited capability:** the optimum or achievable reference under explicitly stated information and timing assumptions. Only use “optimum” when it is actually proved or exactly solved.
2. **Learning benefit:** performance of a separately trained communication policy minus a separately trained matched local policy at a declared training budget.
3. **Frozen-policy reliance:** performance of the same trained policy under live messages minus performance after a stated communication intervention.
4. **Robustness gap:** difference between a channel-appropriate achievable reference and the learned policy, reported alongside success, collisions, decision time, and communication cost.

For higher-is-better success S , a useful empirical display is:

learning benefit = $S(\text{comm trained}) - S(\text{local trained})$

reliance = $S(\text{same policy, live}) - S(\text{same policy, intervention})$

reference gap = $S(\text{reference under this information/channel}) - S(\text{learned policy})$.

A severed policy has an altered input distribution; its drop can reflect brittleness. A negative learning benefit can reflect optimization difficulty. Neither result alone establishes the task's information requirements. Your code already distinguishes much of this; the new work is to connect the measurements to a controlled attainable reference and a useful deployment question.

7.5 What would constitute a contribution

A plausible first paper would need all of the following:

- A precisely specified task family and an elementary certified calibration subcase.
- A controlled experiment separating private-information availability from memory and actor capacity.
- At least one strong recent baseline relevant to the actual claim.
- A reproducible finding about the discrepancy between learned reliance and usable information under timing/failure shifts.
- A robotics-like validation with meaningful movement and a domain endpoint.
- A reusable evaluator or task implementation, seed-level data, and clearly scoped conclusions.

A new algorithm is optional. A new scientific insight is not. “Messages arriving after the deadline do not help” is a sanity check, not a paper. “A claimed efficient communicator's measured advantage reverses under a controlled timing condition for a particular explainable reason, and a simple baseline closes the avoidable gap” could be a contribution if well demonstrated and not already settled by the closest work.

8. A tractable experiment plan

8.1 Stage the study instead of crossing every factor

Stage 0: engineering and analytic checks. Validate the information contract, reward timing, oracle decision rule, independent channel randomness, reset behavior, and no-message ceiling. These are correctness checks; they do not require a large training sweep.

Stage 1: small learning pilot. Train a local recurrent control, a capacity-matched recurrent local control, and dense recurrent communication on development seeds in both information variants. Use a modest fixed pilot allocation. Confirm that the oracle can perform the task and determine whether the learning setup has a usable signal. A failure is retained and diagnosed; do not silently discard seeds or keep changing test tasks until communication wins.

Stage 2: main comparison. Freeze the task version, primary endpoint, methods, training allocation, and held-out evaluation banks. A reasonable planning allocation is two information variants, six methods, and eight independent training seeds: **96 trained policies**. Eight seeds is a starting budget, not proof of sufficient precision. Use pilot variance and a chosen practically meaningful effect to revise the sample plan before opening final outcomes.

Stage 3: deployment evaluation. Evaluate frozen policies across a small fixed set of delay and outage conditions. This can answer deployment-sensitivity questions without retraining every policy under every channel. It does not establish optimal adaptation to each test channel.

Stage 4, optional: train a small subset with channel randomization to ask whether the failure can be learned away. Budget this separately. Do not quietly mix clean-trained and robust-trained results into one ranking.

8.2 Six primary method slots

Slot	Method	Purpose
1	Recurrent local policy	Strong no-message control.
2	Recurrent LocalCapacity policy	Separates remote information from extra active computation.
3	Recurrent dense Broadcast	Simple full-message reference.
4	Recurrent periodic/round-robin communication	Strong practical matched-budget schedule without learned admission.
5	A faithful recent baseline chosen from SLIM or a delay-aware method	Prevents the paper from comparing only with your own older modules. Choice follows the final claim.
6	A transparent temporal baseline or a justified proposed modification	Last-message cache plus age threshold, or one mechanism derived from a demonstrated failure. Do not claim this standard baseline as novel.

Keep a feedforward local policy as a targeted memory diagnostic, with a separately stated allocation. Keep the oracle-information controller as a calibration/evaluation reference, not an unfair trainable competitor. If

the recent baseline requires a very different training method, report that difference rather than forcing an inaccurate reimplementations into the shared shell.

If a faithful external baseline is infeasible, narrow the paper's claim to a controlled analysis of the methods actually evaluated. Naming your modules after classic ideas does not make them exact reproductions of CommNet, IC3Net, TarMAC, or DGN.

8.3 Restrict the channel grid

Start with three timing conditions: on time, near the decision boundary, and too late. Express delay in environment steps and relative to the deadline. This is more interpretable than arbitrary milliseconds without a physical simulator clock.

For a second axis, use clean delivery, iid erasures, and burst erasures with the **same long-run mean delivery rate**. Use this comparison only in the repeated-transmission robot task, with a declared retry schedule and decision window. The single-shot calibration task cannot distinguish iid from burst loss at an identical on-time delivery probability. Report both marginal packet delivery and the probability of at least one useful delivery before commitment; their difference is part of the mechanism. This distinguishes outage structure from average bandwidth. Fix the attempted transmission schedule and payload size across these conditions as well as the marginal delivery rate. Fix the burst process and random seeds before evaluation. Do not add routing, congestion, quantization, adversarial corruption, node failure, and changing team size simultaneously.

The recommended first version uses a fixed communication graph and an exogenous schedule. This avoids claiming a deployed decentralized scheduler while silently using all-agent scores. If learned scheduling becomes central, count requests, keys, acknowledgments, and timing information or explicitly state which are idealized.

8.4 Evaluation and statistical plan

Use success before the deadline as the primary endpoint for the new task. Report navigation success, collision frequency, time to correct commitment, environment return, and actual communication cost as secondary outcomes. For power-grid work, retain physical units and separate voltage quality, reactive effort, losses, and solver failure.

Average evaluation episodes within each trained policy before drawing conclusions across training seeds. More evaluation episodes reduce environment noise; they do not create more independently trained policies. Pair methods by declared training seeds and evaluation conditions, but do not assume identical seed labels make the full learning trajectories identical.

Use a held-out bank large enough to stabilize the primary success estimate; 256 episodes per policy/condition is a reasonable starting allocation for this small task, to be checked against pilot variance. Keep training/development seeds and final evaluation seeds disjoint. For any constructed deterministic calibration case, report its exact arithmetic instead of attaching a population confidence interval to hand-selected states.

Show every seed, paired effect estimates, and uncertainty. A seed-level bootstrap is a descriptive tool, not immunity from small-sample limitations. Predeclare a small number of primary contrasts; label the remaining grid exploratory. A practical effect threshold might be five percentage points of deadline success, but choose it with the advisor based on the application before final analysis. An interval crossing zero does not establish equivalence; a no-difference claim needs an appropriately powered equivalence/noninferiority design and margin.

For multiple tasks, show per-task effects and a carefully chosen aggregate only if the scale and interpretation are meaningful. Do not pool raw Simple Spread return with voltage violation or capture probability. Avoid percent improvement based on negative returns. Report failed runs and incomplete cells instead of calculating only over successful survivors. [Statistical evaluation reference: Agarwal et al., NeurIPS 2021](#)

8.5 Compute and storage planning

The repository's older Simple Spread ledger reports about 43 GPU-hours for 224 completed 600k-frame runs, roughly 0.192 GPU-hours per completed run. Multiplying by 96 gives about 18.4 GPU-hours **only under that historical per-run average**. This is not a forecast for recurrent actors, longer sequences, different environments, or additional evaluation. MAPDN also has a different throughput profile.

Profile one representative new run before reserving the full allocation. Estimate:

```
training worker-hours = number of runs × median measured run-hours
evaluation hours = policies × conditions × episodes × episode runtime.
```

Then add an explicit operational margin, for example 30%, as a planning assumption. Reuse your existing job manifests and artifacts. Tiny actors may be faster on CPU; benchmark representative CPU and GPU configurations without changing the scientific protocol mid-comparison. Independent runs can parallelize, but wall-clock timings from contended jobs are not clean method-speed comparisons.

Archive the source revision, resolved configs, dependency/runtime records, seed list, final policies, seed-level results, and figure-generation code. Preserve raw trajectories needed for the central claim. You already have extensive preservation infrastructure; the next task is to make the evidence accessible and compact, not create another layer of bookkeeping for its own sake.

9. Implementation work mapped to the current repository

This is a proposed implementation sequence, not a request to replace all existing code.

Work package	Existing files to extend	Concrete acceptance check
Current source repair	scripts/run_main_direction_phase.py; src/commstudy/experiments/bookkeepin g.py	A serialized closure record reaches evaluation without a string/Path exception; historical results remain separately attributable.
Current scientific index	README.md; docs/; latest agents.md findings	A new reader can identify latest completed results, limitations, and artifact locations in five minutes.
New versioned task	src/commstudy/tasks/vmas/scenarios/; task registry and configs	Reset distributions, cue privacy, observation masks, deadline, and oracle behavior match the written task contract.

Work package	Existing files to extend	Concrete acceptance check
Recurrent actor	src/commstudy/models/model.py; model config/schema	Hidden state resets on individual environment completion; sequences preserve temporal order; local and communication controls have comparable memory.
Recurrent evaluation	analysis/saliency.py; analysis/diagnostics.py; task-specific evaluators	Paired rollouts start with equivalent state; each intervention evolves its own hidden state; exact local nulls hold.
Temporal delivery	communication/channel.py plus a small explicit delivery-state module	Send time, arrival time, age, validity, and per-environment queue reset are observable in a tiny deterministic trace; no future packets arrive early.
Channel bookkeeping	experiments/bookkeeping.py; metrics and evaluation	Delivered and emitted payloads are distinct; stale/dropped packets and burst state are retained.
Analysis	Existing aggregate/report tools and a dedicated new study analyzer	Primary contrast, every seed, failed cells, information reference, and learning/reliance curves reproduce from saved artifacts.

The highest implementation risk is temporal replay. Current sender-mask replay handles a within-step random channel. A delay queue introduces inter-step state. A mutable queue must not advance whenever PPO reevaluates a minibatch. Receiver histories must not accidentally contain future packets. If past learned payloads are replayed as detached fixed observations, gradients and the sender's credit assignment differ from an end-to-end differentiable temporal communicator; name that design explicitly.

First validate recurrent actors and controls under instantaneous communication. Then freeze those policies and study evaluation-time delay with a fully specified delivery wrapper. Only afterward consider training through delayed communication using sequence-aware collection and updates that preserve hidden-state and realized-channel semantics. Existing feedforward policies can supply an earlier engineering smoke check, but they do not replace the recurrent controls in the main evidence. Do not remove the current evaluator's RNN rejection just to make a configuration run. An external validated recurrent BenchMARL model may be reused, but the project's paired intervention path still needs verification.

Do not add QMIX merely to appear broader. A second algorithm is useful when the paper claims the result is algorithm-independent; it is optional for a focused MAPPO study whose scope is explicit. Likewise, a Transformer, role embeddings, an LLM critic, or a new optimizer should appear only if a concrete hypothesis requires them.

10. What the larger pivots would actually involve

10.1 Resilient cooperative sensing: strongest alternative

Question: how should a robot team allocate sensing and communication so that a shared environmental estimate remains useful through intermittent connectivity?

A feasible task is planar tracking or field mapping with a small team, known sensor noise, local measurements, a limited communication graph, and a fixed mission duration. Use estimation error and energy/travel cost as outcomes. Compare periodic sharing, innovation-triggered sharing, dense communication, local-only estimation, and one learned policy. A Gaussian-process or simpler state estimator provides a meaningful non-MARL reference.

Your likely contribution would need to concern a specific failure pattern or coordination decision: for example, whether delayed measurements should change who explores versus who relays. This is more substantive than adding communication to coverage reward. Penn's Douro and MOCHA work make the application connection credible, but they also raise the baseline bar. The main risk is spending weeks building an estimator and simulator before learning whether there is an interesting coordination problem.

First go/no-go: a scripted estimator/controller must show that exchanging information changes the physical estimation objective; a simple event-triggered baseline must leave measurable room for improvement. If it already solves the problem, a learned architecture is not automatically justified.

10.2 Team changes and role recovery

Question: when an agent fails or an unfamiliar replacement joins, can a team recover useful role allocation without retraining the entire policy?

This fits your earlier interest in heterogeneous roles. Use a controlled scout/responder or cooperative transport task, define failure time and lost capability, and evaluate recovery time and remaining-task success. Distinguish fewer agents, damaged sensing, unfamiliar policies, and changed role composition; these are different shifts.

The current flat observation encoder is not automatically suitable for variable team sizes. Add an entity/set encoder and masks, verify permutation behavior, and separate parameter sharing from true population invariance. Hold out partner policies or role combinations during training. **HyperMARL** and the September 2026 **JaxAHT** preprint are relevant baselines/infrastructure. Reimplementing all of them is unnecessary, but the claim must state which kind of generalization is actually tested. [HyperMARL · JaxAHT, September 12, 2026 preprint](#)

A concrete candidate contribution is understanding whether a communication protocol remains usable after role reassignment, rather than merely demonstrating a higher average score with more agents. This option has good reuse but needs a careful partner-training and evaluation split.

10.3 Safety under communication failure

Question: can a decentralized team maintain a specified safety constraint while degrading gracefully when remote information becomes unavailable?

Choose one tractable constraint, such as pairwise collision avoidance for bounded-speed planar robots. A model-based safety filter or control barrier formulation is a meaningful baseline; a policy trained with a collision penalty is not a safety certificate. Define the state estimate, disturbance bounds, sensing assumptions, and what happens if the feasible safe-action set is empty.

DGPPO already combines graph structure, PPO, and safety-related objectives, and GCBF+ is a serious decentralized safe-control reference. A stronger contribution would address an explicit missing-information assumption or degradation mode, with a proof only where its conditions genuinely hold. If Enyioha wants to supervise the control-theory component closely, this could become the strongest long-term robotics direction. It is a riskier short-term pivot. [DGPPO](#) · [GCBF+](#)

10.4 Grid control and topology transfer

Question: when does inter-zone communication improve voltage regulation beyond local measurements and local optimization, and does the result survive a change in feeder or operating conditions?

You have useful domain infrastructure, but current MAPDN observations already include zone-level measurements, and remote electrical changes propagate into local voltages. “Remote disturbance” does not imply “unobservable local consequence.” A power-flow-aware observation analysis is needed before asserting private information.

A defensible study would compare against local droop/Volt-VAR or another appropriate engineering controller and an optimization reference, define inverter limits, and report violation magnitude/frequency, reactive effort, losses, and solve failures. Electrical adjacency and radio/network connectivity must be distinct objects. One feeder with new time windows supports temporal evaluation on that feeder, not general topology transfer.

This could align very well with your advisor and broader power-grid work, but it is less directly a robotics thesis. Keep your separate multi-agent reasoning system for datacenter interconnection distinct from MARL unless you explicitly introduce a learned sequential decision problem. Connecting two projects because they both have “agents” is not itself a research contribution.

10.5 Offline or continual coordination

Offline MARL would require a dataset with meaningful behavior diversity and support for the actions you want to evaluate. Your current checkpoints/trajectories could seed a dataset, but they may cover only a narrow set of successful behaviors. Off-policy evaluation and dataset construction could consume the project.

Continual coordination is more plausible as a longer-term extension: study whether changing one agent's policy or communication encoder breaks compatibility with partners. However, **MEAL already studies coordination forgetting and partner changes**. A specific protocol-compatibility question, delayed teammate updates, or asymmetric retraining regime would be needed. **Oryx** provides a modern offline MARL reference. [MEAL](#) · [Oryx](#)

10.6 Why I would defer an LLM or world-model thesis

You have enough software experience to build a compelling demo, but a compelling demo is different from a clean MARL contribution. Adding an LLM creates questions about pretrained knowledge, prompt sensitivity, inference cost, reproducibility, and whether learning actually occurs across agents. A large critic can also introduce privileged information and expensive training without isolating a useful scientific mechanism.

A bounded side experiment could use a frozen planner to generate a high-level task allocation, then compare it to a simple heuristic using the same observations and cost accounting. I would not make that the main thesis unless the advisor identifies a sharply scoped research gap and you deliberately choose to abandon much of the current experimental investment.

11. A concrete schedule

This schedule starts the week of September 21, 2026. It assumes the part-time research allocation above. Your actual honors-thesis and graduate-application deadlines were not verified; synchronize this schedule with those dates rather than treating it as an institutional calendar.

The first ten working days

Days	Work	Tangible output
1-2	Reconcile current experiment results; correct observation/full-state claims; preserve historical reports	A two-page current evidence memo and an accessible artifact index.
2-4	Read Probing, SLIM, CoDe/CAIC, and the relevant utility-scheduling papers; compare with the exact proposed question	A one-page nearest-work matrix with a specific unresolved claim.
3-5	Meet Enyioha about the research question, scope, target program/lab, and time budget	One primary question, one fallback, and a first-paper contribution type.
5-7	Specify the task contract and analytic calibration; implement oracle/local decision checks	A small functioning calibration task with verified no-leakage assumptions.
7-9	Profile the learning path and run only development pilots	One figure showing task solvability and whether learning can exploit the controlled information.
10	Review novelty and feasibility together	Continue, narrow, or pivot based on explicit evidence.

Continue if: the literature comparison identifies a distinct question; the oracle solves the task; the information controls behave as specified; and there is a plausible learning/evaluation path within the available time. A learned communication pilot that fails does not invalidate the information proof, but it signals that the proposed learning study may be too expensive without a better-supported baseline.

Narrow if: timing/recurrent replay is taking over the project. A clean memory-versus-communication study with a controlled task can be preferable to a broken network simulator.

Pivot to option 2 or 3 if: the closest work already answers the proposed question, or the new task remains an artificial signaling exercise with no useful robotic consequence. Do not extend the pilot indefinitely just to protect the initial choice.

The following twelve weeks

Period	Scientific milestone	Required evidence
Weeks 1-2: Sep 21-Oct 4	Novelty and task feasibility	Direct predecessor comparison; analytic/oracle controls; bounded pilot.
Weeks 3-4: Oct 5-18	Recurrent controls and delivery/evaluation correctness	Sequence resets, replay semantics, no leakage, exact local nulls, representative throughput.
Weeks 5-6: Oct 19-Nov 1	Main training and first held-out analysis	Frozen methods/budget/seeds; complete cohort; primary endpoint with every seed.
Weeks 7-8: Nov 2-15	One targeted robustness/generalization test	New predefined channel/layout conditions; explanation of failure relative to reference.
Weeks 9-10: Nov 16-29	Paper draft and reproducible release candidate	Introduction, related work, methods, primary figures, limitations, clean reproduction path.
Weeks 11-12: Nov 30-Dec 13	Advisor review and focused revisions	Resolve specific reviewer concerns; polish evidence and presentation.

This is a minimum coherent-paper schedule, not a promise of acceptance within twelve weeks. If graduate applications occur earlier, prepare a truthful interim research statement with preliminary results and a clearly labeled manuscript in preparation. Do not call an unfinished draft submitted or accepted.

12. What to write, show, and bring to your advisor

12.1 Suggested paper argument

A strong manuscript could follow this sequence:

1. **Problem:** message use, learned benefit, and task information requirements are often conflated in empirical comparisons.
2. **Existing evidence:** your current controlled results show that adding communication does not reliably help, despite substantial infrastructure and training.
3. **Experimental construction:** define a task family with controlled information availability and commitment timing, including an analytically transparent subcase.
4. **Measurements:** separate attainable references, retrained baselines, frozen-policy interventions, and communication cost.
5. **Main finding:** report the actual observed mechanism, including negative or contrary outcomes.
6. **Robotics consequence:** explain the resulting sensing, communication, or fallback decision in the continuous-control setting.
7. **Limits:** scope the claim to policy classes, budgets, channel models, and environments tested.

Do not make the introduction a list of all modules in the repository. The contribution is the answer to a research question. The software supports that answer.

12.2 Five figures worth producing

- **Information contract and task geometry:** who knows what, when a decision becomes irreversible, and which channels are available.
- **Capability and achieved performance:** oracle/reference curve and learned methods as timely delivery changes.
- **Memory and capacity controls:** paired-seed effects for feedforward/local recurrent/active-capacity/communication policies.
- **Reliance versus benefit:** each trained policy's intervention effect compared with its independently trained local comparator.
- **Held-out robustness:** matched mean delivery but different outage structure, or one carefully chosen external task.

These should be generated from actual results. A large architecture diagram or a montage of robot motion is supplementary unless it makes the scientific mechanism easier to understand.

12.3 The graduate-research portfolio

Aim to have a readable manuscript or preprint, a concise project page or repository landing page, a short video explaining the experiment, and one command that reproduces a small result. Keep the complete artifact packet accessible. Clearly identify your own contributions: task formulation, implementation, experimental design, mathematical analysis, empirical finding, and writing.

The advisor letter can then discuss how you responded when the original prediction was not supported: diagnosing the task, distinguishing alternatives, designing stronger controls, and producing a defensible conclusion. That is a more credible research narrative than claiming every experiment succeeded.

For venue planning, discuss AAMAS if the core is multi-agent communication/evaluation; CoRL or a relevant robotics venue if the physical learning contribution is strong; and ICRA/IROS if the robot-system consequence is sufficiently grounded. A workshop can be a useful early audience, but call it a workshop. These are scope-fit suggestions, not acceptance predictions, and conference deadlines were not verified here.

12.4 Questions for Enyioha

1. Is the intended contribution primarily empirical methodology, a communication/control method, or an application result?
2. Does the proposed information-and-deadline question remain distinct after reading the closest 2026 work?
3. Would a calibrated benchmark plus a clear robotic finding be enough, or should the paper require an algorithmic advance?
4. Is cooperative sensing a stronger application than the current prey-capture task for this lab's expertise?
5. What is the actual weekly research allocation and final thesis deadline?
6. Can the lab support a small physical demonstration, and would it materially test the claim?

7. Is Pappas/ESE the intended Penn lineage, or is AMCS a deliberate preference tied to another advisor?
8. Which completed negative results belong in the main paper, and which should remain a separate technical appendix?

13. Reading order with a purpose

Do not read every paper cover to cover before doing anything. Read the first tier closely enough to write the novelty matrix; use the others when the corresponding direction becomes relevant.

Order	Paper	What to extract
1	Probing Dec-POMDP Reasoning, AAMAS 2026	What its diagnostics establish, what they cannot establish, and how reactive/recurrent controls are used.
2	SLIM, 2026 preprint	Actor-capacity separation, budget definitions, task settings, and network effects outside its scope.
3	CoDe, AAAI 2025	How delayed messages are represented and fused; what counts as a fair delay-aware comparator.
4	CAIC, UAI 2026	Queueing assumptions and the exact relationship between communication load, age, and performance.
5	SCoUT, 2026 preprint	Counterfactual communication credit, decentralized execution, and multi-message interaction limits.
6	COBE, AAMAS 2026	What a cost-benefit framing already covers.
7	SeqComm-DFL, 2026 preprint	Necessity/value assumptions; check the proof rather than importing its claims.
8	Pitfalls of Measuring Emergent Communication, AAMAS 2019	Signaling, influence, and functional communication distinctions.
9	Deep RL at the Edge of the Statistical Precipice, NeurIPS 2021	Seed-level uncertainty, comparison quality, and appropriate aggregate reporting.
10	Douro River Plume MARL, ICRA 2026	How target-lab work motivates a concrete mission and evaluates physical utility.
11	MOCHA, ICRA 2024	What real intermittent communication requires beyond iid dropout.
12	DGPPPO, ICLR 2025	Safety assumptions and comparison requirements if choosing the control pivot.
13	HyperMARL, NeurIPS 2025	Agent diversity and parameter-sharing alternatives for the team-change pivot.
14	JaxAHT, September 2026 preprint	Modern ad hoc-team evaluation and infrastructure, if open teams become central.

Order	Paper	What to extract
15	MEAL, ICML 2026	Existing continual-coordination questions, to avoid proposing them as new.

For each of the first seven, write five lines: problem, information assumptions, training/execution model, evaluation coverage, and the exact question left unresolved. That small document is more valuable than a large folder of unread PDFs.

14. Final decision rules and limits

Retain: MAPPO/BenchMARL, communication modules as baselines, active local controls, paired interventions, frozen result history, and the existing physical metrics.

Change next: the precision of the research question, the task's decision-relevant information contract, the strength of local memory controls, and the interpretability of the experimental outcome.

Defer: a full four-mechanism MAPDN sweep, QMIX for breadth alone, a new GNN architecture without a mechanism, a multi-domain benchmark expansion, and an LLM/world-model rewrite.

The highest-value immediate deliverable is a one-page claim with one decisive experiment, supported by the nearest-work matrix and a working positive-control task. If that claim survives scrutiny, the existing repository gives you a credible foundation to finish it.

The research search was broad but not exhaustive. Some pages were inaccessible; accepted venue labels are used only where supported by an inspected primary record, and other work is labeled preprint. Faculty recruitment, AMCS admissions details, institutional deadlines, and application outcomes remain unverified. Repository results remain documentary evidence until the retained raw artifacts are independently available. No new numerical research result or novelty guarantee is claimed in this report.